

Conventional NIRS calibration

external pipeline search

many preprocessing pipelines

CV selects a complete pipeline

final model depends on pipeline choice

external choices are hard to audit

refitting repeats the search

Analytical bottleneck

cost, variance, opacity

high calibration cost

winner's curse & instability

deployment complexity

poor traceability

limited routine usability

Operator-adaptive calibration

model-internal & auditable

compact linear operator bank

selection inside the calibration model

original-wavelength coefficients

selected operators logged

fast refit of one linear model

AOM reframes preprocessing selection as a model-internal, auditable part of analytical calibration.